

Artificial Intelligence Project Pitch

Presented by Group 02

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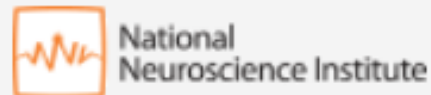
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INTRODUCTION

SINGHEALTH'S BACKGROUND

- Largest healthcare group in Singapore since 1998
- Operations across public hospitals, community hospitals, specialty centers, and polyclinics



SINGHEALTH'S VISION

- Continuous Modernization and Community Health Enhancement
- Emphasis on Digitizing Patient Records and Integrating AI

CYBER INCIDENTS IN 2018

- Negative Impact on SingHealth's Reputation

SWOT ANALYSIS

STRENGTHS

- Reputation
- Robust Infrastructure
- Skilled workforce
- Research and Innovation

WEAKNESSES

- Reliance on Government Funding
- Limited human capability in assessing and responding to risks.
- No Proper Network Security Team

OPPORTUNITIES

- Technological Advancement
- International Collaboration

THREATS

- Regulatory Changes
- Cybersecurity Risks
- Competitors

PESTLE



POLITICAL

- Huge Support and Control from Government Policies
- Political Stability



ECONOMIC

- Healthcare Expenditure
- Economic Growth



SOCIAL

- High Community Awareness of Personal Information
- High AI Readiness in Government, Business, and Consumers (Salesforce 2021).



TECHNOLOGY

- Technology Advancement
- Data Management and Privacy



LEGAL

- Resource consumption (hardware, software, etc.)



ENVIRONMENTAL

- National Healthcare Regulations (HIPAA;...)
- Intellectual Property Protection (Förster n.d.)
- Domestic Regulations (PDPA)



Problem Statement



“Data privacy” is no longer a nice-to-have but turned into “data security” - a must.



Cyber attacks damaged reputation and caused major lost (Jalali & Kaiser 2014).



SingHealth weaknesses cannot be addressed with human efforts ==> require the assistance of automated machine.

Related AI Solution



CyberAI based on ML and NLP technology

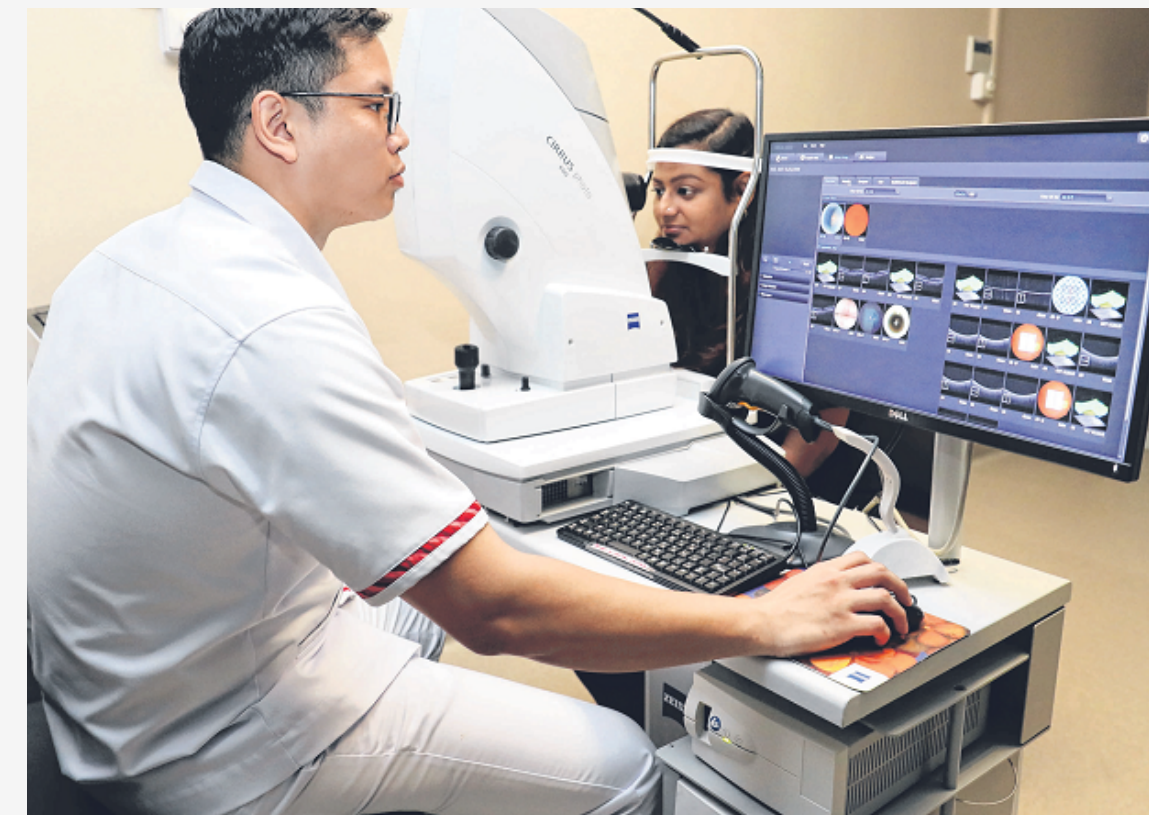
Current AI applications



Doctor Covid

A chatbot to translate medical conversations.

Technology: NLP, ML



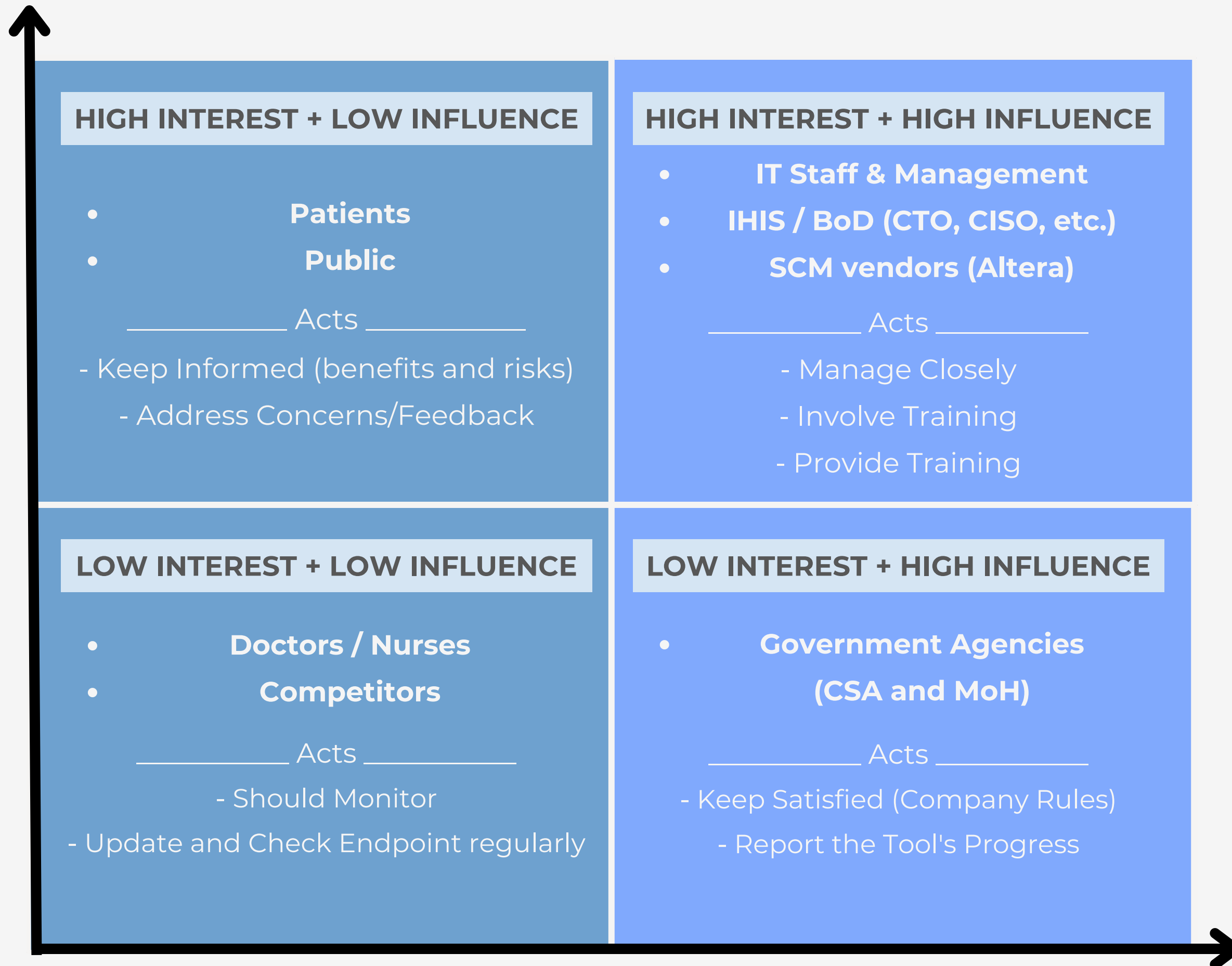
SELENA+

Singapore Eye Lesion Analyser, image reader that can analyse eye scan for detecting diabetes.

Technology: NLP, ML

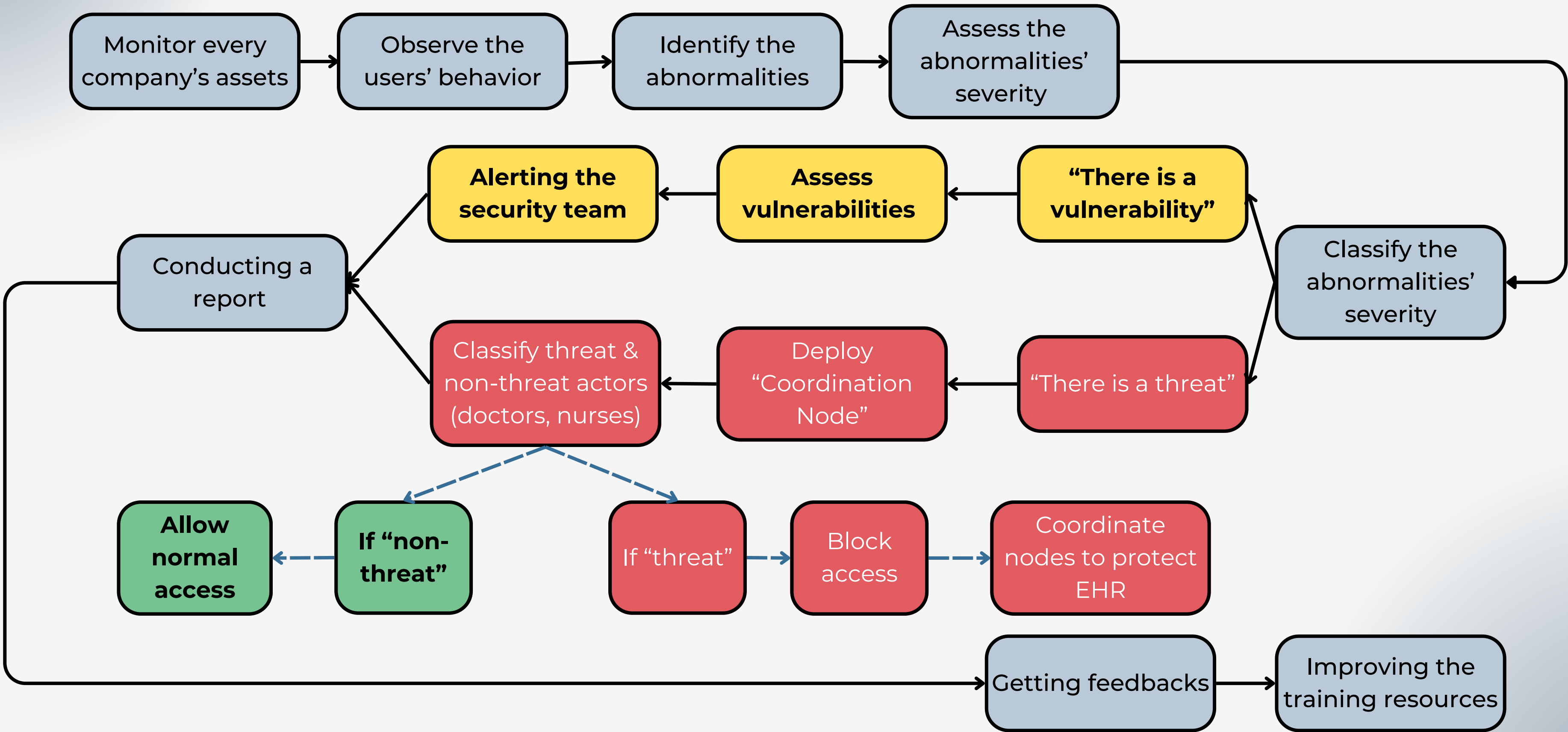
Stakeholders

Analysis

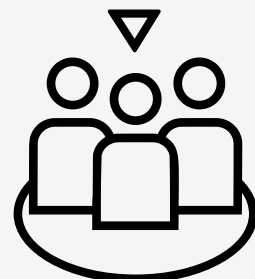


AI Strategy for CyberAI

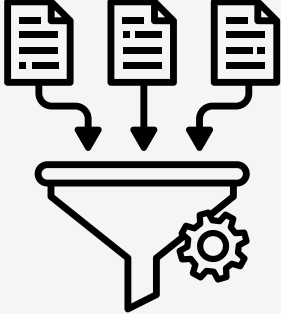
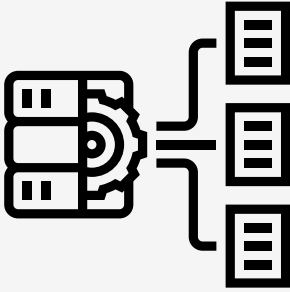
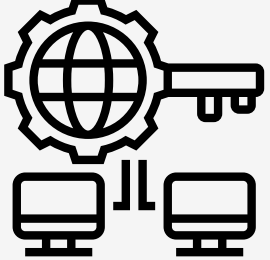
AI step-by-step process



Data strategy

	Data Category	Collecting Methods	Processing Methods	
	The System, Network, Endpoints Behavior	SIEM system EDR, IDS/ IDP (Vectra n.d.)	Unsupervised Machine Learning	
	Upcoming threats	Cybersecurity reports Hackers' forum	Predictive Analysis	NLP
	Users' Behavior	Cookies (Marr 2017)		Behavioural Analysis
	Common threat characteristics	IDS/ IPS		Supervised ML
	Authorized accounts access list	SIEM system	Supervised Machine Learning	

Architecture

Layers of big data	Infrastructures (Tools/Services/Technology)
 Data collection	• SIEM system (ManageEngine Log360) • EDR agent (Cisco AMP) • IDS/IPS (Cisco Secure IPS) • Cookies
 Data storage	The Sunrise Clinical Management (SCM)
 Data analysis and processing	TensorFlow
 Providing access to data	Citrix Virtual Application

 AMP for Endpoints

ManageEngine
Log360

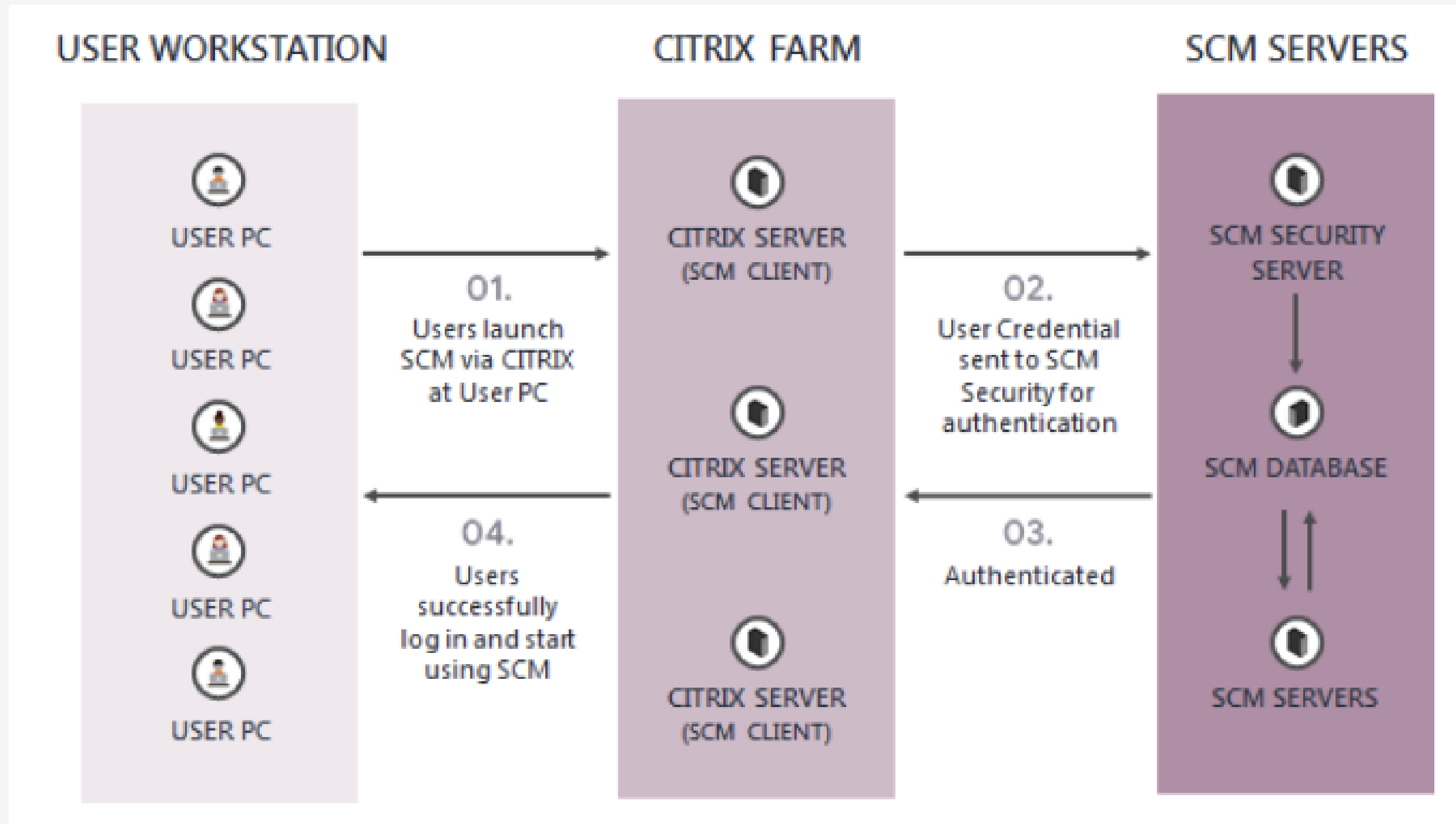

CISCO


SUNRISE
HEALTHCARE
MANAGEMENT

 TensorFlow
2.0



“Data Storage” - “Providing access to data” Layers



Source: COI (2019)

Data Transmission

ABOUT HEALTHCARE CLOUD (H-CLOUD)

Healthcare Cloud (H-Cloud) is a consolidated, cloud computing platform supporting healthcare staff at all public hospitals, eight specialty centres, polyclinics and nursing homes, to retrieve and access patient records.

H-Cloud is the public healthcare's first private cloud setup that reduces operational costs and improves infrastructure availability to 99.95%.

H-Cloud is ISO 20000-1 (Information Technology Service Management) and 27001 (Information Security Management System) certified. These international certifications help to assure all stakeholders that H-Cloud processes follow industry best practices



Enhanced operational efficiency and higher resiliency



Cost savings for healthcare institutions and patients



Well-equipped to defend and detect cybersecurity threats

Source: Synapxe (n.d.)

Organizational capabilities

Strengths

- Fully Digitized Dataset
- Strong Research Culture
- Hospital Budget

Weaknesses

- IT Staff Mindset
- Cybersecurity Expertise of Senior Management level

Change Management



Ensure buy-in from all stakeholders

Effectively communicating the benefits of AI brings to cyber security, such as enhanced threat detection and mitigation



Delineation of responsibilities

Clearly defining which are tasked with AI model training, system integration, and ongoing AI maintenance



Training programs for appropriate employees

Ensuring that all employees understand how to effectively use the AI system



Long-term strategic planning of AI solutions

Regular system updates, SingHealth must provide ongoing training opportunities for staff

BUSINESS MODEL CANVAS

KEY ACTIVITIES

- Empathise SingHealth problems and develop a customized AI software solution.
- Help deploy the software integrated with SingHealth's current system.
- Train security staff to adapt to the new software.
- Maintenance and keep the software up-to-date.

KEY PARTNERS

- Government agencies - must comply with and receive support from.
- Vendors/ Suppliers of high-quality software/hardware.

KEY RESOURCES

AI Platform powered by data on phishing behaviour collected by Google

COST STRUCTURE

- **Fixed costs:** Proof-of-Concept, Infrastructure, Cloud server, Professionals.
- **Variable costs:** Extra cloud storage if data increases.

VALUE PROPOSITION

An AI-based solution that transforms the way SingHealth stores and protects customers' data.

CUSTOMER RELATIONSHIP

Offer 60-day free trial for an MVP.

CHANNELS

- **AI Platform:** TensorFlow.
- **Cloud:** H-cloud.

REVENUE STREAMS

- **Software sales**
- **Support and Care** service, including deployment and frequent maintenance.

CUSTOMER SEGMENT

Business-to-business: Hospitals.

- **Firmographics:** located in Eastern Singapore, leading the Eastern network of healthcare.
- **Technographic:** highly digitalized, operate collaboratively in a cloud-based suite.
- **Needs-based:** quality-focused segment.
- **Sophistication-based:** never used AI for security before.

FINANCIAL PLAN

COST STRUCTURES

Tasks	Tools/ Infrastructure	Price (\$/month)
Data collecting	IDS/IPS (Cisco Secured IDS)	\$3,000
	SIEM (Log360)	\$200
	EDR (Cisco AMP)	\$1,680
Data storage	EHR database by SCM	\$1,000
	H-cloud	\$1,500
Data analysis and processing	TensorFlow	\$0.00
Providing access to data	Citrix	\$7,000
Developers salary	_ 2 Developers	\$4,000
Project Manager salary		\$3,000
Developers + Data Scientist	_ 3 months	\$4,000
Total		\$25,380

REVENUE STREAMS

Total cost to develop the full application

\$913,680

Price per a unit (exclude other services):

\$70,000

To achieve break-even:

13 packages

Estimate time till break-even:

> 6 months

Concerns



Legal responsibility and accountability

- Designating who is responsible for the actions of AI systems, ensuring that protocols are in place.
- Involves an individual or group developing a comprehensive set of terms and conditions.
- Ex: If the AI erroneously predicts a cyber attack, the AI provider will be held accountable



Incident response and reporting

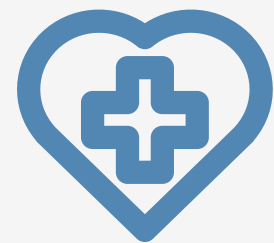
- Building and maintaining an effective and timely incident response plan for all security breaches.
- Ensuring compliance with Singapore's data protection laws.



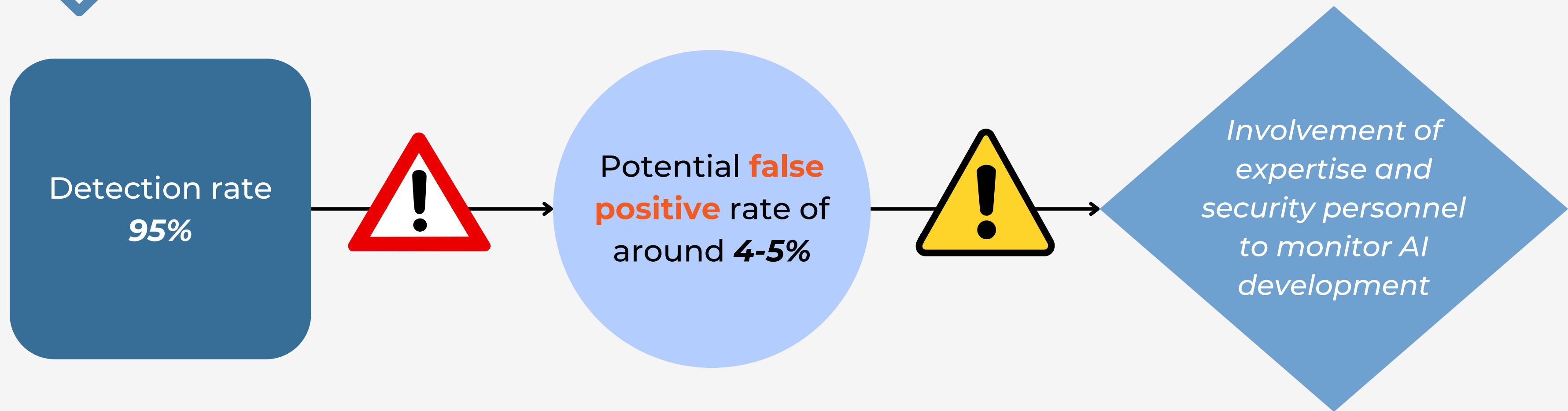
Government regulations and guidelines

- Clear procedures for reporting breaches to authorities and affected individuals. Ensuring compliance with Singapore's data protection laws.
- Monitoring regulatory developments, participating in industry discussions, and seeking legal advice

Concerns



Accuracy



**Thank You
For Your
Attention**



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